

Mensuration Formulas

2D Shape Area Formulas

Rectangle

$$\text{Area} = \text{length} \times \text{breadth}$$

Square

$$\text{Area} = (\text{side})^2$$

Triangle

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Triangle (Heron's Formula)

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where,

$$s = \frac{a + b + c}{2}$$

(semi-perimeter)

Equilateral Triangle

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

Parallelogram

$$\text{Area} = \text{base} \times \text{height}$$

Rhombus

$$\text{Area} = \frac{1}{2} \times d_1 \times d_2$$

Trapezium

$$\text{Area} = \frac{1}{2}(a + b)h$$

where:

- a and b are parallel sides
- h is height

Circle

$$\text{Area} = \pi r^2$$

(r = radius)

Circle Sector

$$\text{Area} = \frac{\theta}{360^\circ} \pi r^2$$

(θ is the angle in degrees)

Area of a Ring/Path

$$\text{Area} = \pi(R^2 - r^2)$$

where:

- R = Outer radius
- r = Inner radius

Important Perimeter Formulas

Rectangle

$$\text{Perimeter} = 2(l + b)$$

Square

$$\text{Perimeter} = 4a$$

Circle (Circumference)

$$\text{Circumference} = 2\pi r$$

Semi-circle Perimeter

$$\text{Perimeter} = \pi r + 2r$$

3D Shape Surface Area Formulas

Cube

$$\text{Surface Area} = 6a^2$$

Cuboid

$$\text{Surface Area} = 2(lb + bh + hl)$$

Cylinder

Curved Surface Area (CSA)

$$2\pi rh$$

Total Surface Area (TSA)

$$2\pi r(h + r)$$

Cone

Curved Surface Area (CSA)

$$\pi r l$$

Total Surface Area (TSA)

$$\pi r(l + r)$$

where:

$$l = \sqrt{r^2 + h^2}$$

(slant height)

Sphere

$$\text{Surface Area} = 4\pi r^2$$

Hemisphere

Curved Surface Area

$$2\pi r^2$$

Total Surface Area

$$3\pi r^2$$

Key Tips for Aptitude

- Always check that units are consistent before calculation.
- Convert cm to m or vice versa when required.

Diagonal Formulas

Diagonal of Square

$$d = a\sqrt{2}$$

Diagonal of Rectangle

$$d = \sqrt{l^2 + b^2}$$

Area of 4 Walls

$$\text{Area} = 2h(l + b)$$

where:

- l = length
- b = breadth
- h = height